



Assessing uncertainty in VGI for emergency response

Michael E. Camponovo & Scott M. Freundsuh

To cite this article: Michael E. Camponovo & Scott M. Freundsuh (2014) Assessing uncertainty in VGI for emergency response, *Cartography and Geographic Information Science*, 41:5, 440-455, DOI: [10.1080/15230406.2014.950332](https://doi.org/10.1080/15230406.2014.950332)

To link to this article: <https://doi.org/10.1080/15230406.2014.950332>



Published online: 02 Sep 2014.



Submit your article to this journal [↗](#)



Article views: 930



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 7 View citing articles [↗](#)

Assessing uncertainty in VGI for emergency response

Michael E. Camponovo^a and Scott M. Freundsuh^{b*}

^a*Earth Data Analysis Center, University of New Mexico, Albuquerque, New Mexico, USA;* ^b*Department of Geography and Environmental Studies, University of New Mexico, Albuquerque, New Mexico, USA*

(Received 19 February 2014; accepted 21 July 2014)

This research project examines the geographic data produced by volunteers via the Ushahidi web platform in response to the earthquake that struck Haiti in January 2010. Volunteers translated messages (text, e-mail, and voice) submitted by victims of the earthquake, categorized each message into a primary ‘emergency need’ category and subcategory, and georeferenced each message on a dynamic web-based map. Initial inspection of the categorized data indicated discrepancies between the emergency need submitted by victims and the subsequent categorization of the emergency need. Analysis of the main categorical data illustrated that 50% of the messages were mis-categorized by the volunteers, failing to convey the main idea of the victim’s message. At the subcategory level, approximately 73% of the messages failed to convey the main idea of the messages. These numbers are higher than the estimate of 36% error in categorization produced in an independent review of the Haiti Ushahidi database. While the volunteer response to the Haitian earthquake represents a paradigm shift in emergency response and victim empowerment that has been repeated in numerous natural and man-made disasters around the world, this study suggests the need for more research on the quality of the categorization (i.e., attribute data) of volunteered emergency data.

Keywords: VGI; uncertainty; emergency response; attribute quality; GIS

Introduction

When the 7.0 magnitude earthquake struck Port-au-Prince at 4:53 pm on 12 January 2010, Jens Kristensen at the United Nations Headquarters in Port-au-Prince found himself transformed from someone who provided relief to those in Haiti to a victim needing emergency aid. Mr Kristensen’s third floor office was now at ground level surrounded by piles of debris. Within a few short hours of the earthquake, a small group of volunteers over sixteen hundred miles away in Boston, Massachusetts, set up a web site to collect tweets, text messages, e-mails, and news reports about the disaster and placed those reports on a dynamic web-based map. At exactly 11:01 am the next day, the volunteers received and published the following message for anyone, including relief agencies in Haiti, to view:

Over 100 #UN personnel trapped in collapsed headquarters in #Haiti earthquake... (Ushahidi 2011)

Mr Kristensen waited five days, until the afternoon of Sunday, 17 January 2010, when the Fairfax County Search and Rescue Team pulled Mr Kristensen from the rubble of his former office (LaFranchi 2010).

Since 2005 or so, situations like this have become common occurrences in the wake of natural disasters because communication technologies continue to advance and are becoming more ubiquitous. In disasters that have spanned six continents, everyday citizens are reporting geo-

located information through social media applications such as Twitter. Traditional aid agencies are incorporating this information into their response protocols, oftentimes saving lives in the process. This convergence of technology, social media, and geography have spawned a new area of research called volunteered geographic information (VGI) that enables anyone, regardless of geographic knowledge, to produce geographic data with potentially significant impacts (Goodchild 2007).

Despite VGI facilitating the rapid accumulation of data from numerous human sensors, it is often criticized in the literature for questionable quality (Goodchild and Glennon 2010). The research reported in this article assesses the quality of data submitted by ‘citizen’ geographers in the emergency response phase of disaster situations. *Specifically, we assessed the nature of uncertainties in the volunteer-reported, emergency response data distributed via the Ushahidi geospatial database created in response to the 2010 Haiti earthquake.* Understanding the uncertainty in VGI used in emergency response is necessary because if data are incorrect, the resulting geographic data/distributions contain errors and are therefore an important consideration for decision-makers (Hesse 2010).

Background

Historically, geospatial data have been created by and shared from a select group of organizations commonly

*Corresponding author. Email: sfreunds@unm.edu

referred to as ‘authoritative data sources’. In the United States, agencies like the United States Geologic Survey (USGS) have been responsible for collecting and disseminating geographic information. This relative paucity of data collectors was due to the high costs associated with collecting geographic data.

In recent years, the high costs associated with mapping have significantly decreased. Handheld GPS units are relatively inexpensive and are included on most cellular phones. Open-source and free versions of mapping software have been produced that provide various levels of the functionality present in commercial geographic information systems (GISs). The increase in the availability of broadband Internet access has allowed for the quick and easy sharing of data. By lowering the cost of creating and sharing geographic information, mapping has become part of the Web 2.0 movement that allows not only the use of data shared over the Internet but also the collection of data. This melding of Web 2.0 and geography has facilitated VGI (Goodchild 2007). VGI has many potential benefits for the geospatial community: VGI can provide free access to data instead of relying only on commercial options, it can facilitate faster data updates, and data that were previously too insignificant or costly to collect is easily mappable (e.g., Twitter post) (Bloch and Carter 2012).

While this democratization of geospatial data has many potential benefits, it is not without problems. When authoritative data were collected in the past, the quality of geospatial data was assumed to be relatively high. This was due both to the nature of the provider, oftentimes being government agencies and reputable commercial providers, and to the time and effort that went into collecting and producing data. The notion of assumed quality has declined since almost anyone can now create geographic data, regardless of training or expertise. Recent disasters in Haiti and Libya have shown that organizations that use geographic information are willing to make decisions based on volunteered data that may be of questionable accuracy and credibility (Standby Volunteer Task Force 2011). As more organizations begin to embrace social media, more questions will surely arise over the quality of the information that is being shared by individuals and the impacts that data quality might have on the decisions that the agencies make. Agencies that work in fields that are not greatly affected by time have the luxury to assess volunteered data and determine its worth (Haklay 2010; Haklay et al. 2010; Haklay and Ellul 2010; Girres and Touya 2010; Zielstra and Zipf 2010). But what about those that regularly have to make rapid decisions that can have a profound impact on life or death? Emergency response organizations like the American Red Cross, firefighters, and 911 dispatchers may not have the luxury of time to assess the quality and credibility of the volunteered data that are being

shared with them. If these agencies act upon data of questionable quality, what are the potential impacts? This article touches on these questions by studying the quality of VGI produced during a disaster.

Emergency response

There is a substantial body of literature on defining and assessing the quality of geolocation information from GIS used in disaster management and on the application of GISs in all stages of emergency preparation and response (Cova 1999; Haklay 2010; Liu and Palen 2010; National Research Council 2007a; Radke et al. 2000). It is only recently that research has considered social media and crowdsourcing as a source for geographic information. Research has considered the quality of VGI (Haklay 2010; Haklay et al. 2010; Haklay and Ellul 2010; Girres and Touya 2010; Zielstra and Zipf 2010), and research has explored the role VGI plays in disaster response (Standby Task Force, 2011; Norheim-Hagtun and Meier 2010; Pitzer 2011); however, there is little research on the intersection of *spatial data quality*, *crowdsourcing/social media*, and *GISystems* in disaster response.

GIS and emergency response

Actions taken in response to a disaster comprise the ‘emergency management cycle’. This cycle is typically categorized by four phases: preparedness, response, recovery, and mitigation. The preparation phase occurs prior to the onset of the emergency, the response phase takes place immediately after the disaster, the recovery phase is typified by the actions necessary to return life in the affected community to normal, and the mitigation phase strives to limit the effect of future emergencies on the affected community. For a complete discussion on the emergency management cycle, see Cova (1999), Radke et al. (2000), and the National Research Council (2007a).

GISs can and do play a key role in each phase of the emergency management cycle (Cova 1999; Radke et al. 2000; National Research Council 2007a). For a GIS to be effective, emergency responders need detailed spatial information for the affected region. Optimally, responders would need baseline data like roads, buildings, and infrastructure pre-disaster (National Research Council 2007a; Van Westen and Georgiadou 2001), census and demographic data (National Research Council 2007b), access to post-disaster imagery of the affected region for damage assessment (Kelmelis et al. 2006; National Research Council 2007a), and data pertaining to the locations of movements of people impacted by the disaster (Kaiser et al. 2003). All of these data comprise the spatial decision support system (SDSS) that assists emergency responders’ decision-making in time-critical situations.

Crowdsourcing and social media

Evolving technologies have facilitated the creation and sharing of digital content in what is referred to as Web 2.0 (O'Reilly 2005). These technologies are playing a significant role in the development of VGI, where amateurs can submit and create geographic information just as easily as professionals. The cost of global positioning satellite receivers has decreased over time, while their accuracy and ubiquity have increased. Organizations like Google (<http://maps.google.com>) and Microsoft (<http://www.bing.com/maps>) provide free imagery online as well. Many pieces of software for the creation of maps are now available for free (<http://earth.google.com> and <http://www.google.com/mapmaker>), while sophisticated GIS analysis can be performed without purchasing expensive proprietary software (<http://grass.fbk.eu> and <http://www.qgis.org>). These free technologies have spawned what has become known as neo-geography (Turner 2006). Online users have created social networks based on their common interests in utilizing these tools, resulting in what is today known as crowdsourcing (Howe 2008). Common examples of crowdsourcing are the free online encyclopedia Wikipedia (<http://www.wikipedia.org>) and the open-source Linux operating system (<http://www.linux.com>).

Crowdsourcing has been used in emergency management and disaster response (Liu and Palen 2010). For example, during the Indian Ocean Tsunami of 2004, programmers in Sri Lanka pooled their coding skills and within three weeks of the natural disaster created a free and open-source emergency response software package that was modifiable, scalable, operated with minimal hardware and software, and protected the privacy of users and contributors to the software (Curron, De Silva, and Van De Walle 2007). Their software, named Sahana (<http://sahanafoundation.org>), has since been used in numerous emergency situations around the world. Another application designed for smart phones, Outbreaks Near Me (<http://healthmap.org/outbreaksnearme>), also incorporates crowdsourcing. Users of the app can report locations where people are infected with various diseases, like swine flu, allowing health professionals and the community to monitor the spread of disease (Freifeld et al. 2010).

Another important component of crowdsourcing is online social media. Examples of social media applications include Facebook (<http://www.facebook.com>) and Twitter (<http://twitter.com>) where users can share information with their friends and followers. As the popularity of social media increases, organizations and corporations increasingly rely on these media for information (American Red Cross 2010). Combining social media and crowdsourcing with geography has spawned research on VGI (Goodchild 2007). An example of an application of VGI is the Ushahidi platform (<http://ushahidi.com>) that was developed in Kenya. Kenya was suffering from post-election violence during 2007 and 2008, yet citizens could not learn much about the situation

because of a media blackout. The designer of the software operated a blog and asked her followers to send her e-mails pertaining to the violence they witnessed. She was quickly overrun with reports and asked software developers and programmers to develop a system that would help automate the reporting process. The resulting software collected citizen reports that were categorized and geo-tagged and then placed on an online map. The software is open to anyone to submit a report, and anyone can access the reports (Okolloh 2008). A combination of Ushahidi as well as other VGI platforms played a key role a few years later during the Haiti earthquake of 2010.

Crowdsourcing and social media in Haiti

At the onset of the earthquake in Port-au-Prince, there was little geospatial data for the affected region. Therefore, the first task was to create the base data that first responders needed to make decisions. The Humanitarian OpenStreetMap Team (HOT) (<http://hot.openstreetmap.org/>) began this process by heads-up digitizing streets and buildings from aerial and satellite imagery. However, the volunteers did not know the names of the streets or buildings they were digitizing and therefore were unable to assign name attributes. Haitian expatriates living in Boston were enlisted to assist in labeling the streets and buildings. The HOT team relied on Web 2.0 technology and crowdsourcing to quickly create geospatial data used in the emergency response (Nelson, Sigal, and Zambrano 2010). Ushahidi was used extensively to collect real-time information from affected Haitians on the ground (Norheim-Hagtun and Meier 2010). A special short message system (SMS) was set up that allowed affected citizens to report conditions and request help via texts (Pitzer 2011). For instance, some text messages stated where a person was trapped in a building, or that there was not enough water at a shelter. An initial problem with the texts, however, was that few of the responders and Ushahidi volunteers spoke Haitian Creole, the dominant language in Haiti. A crowdsourcing system was developed to allow Haitian expatriates to translate the text messages to English so that they could be mapped, coded, and then acted upon by relief agencies (Munro 2010). Other volunteers mined Twitter and Facebook for updates related to the emergency and mapped those on Ushahidi (Nelson, Sigal, and Zambrano 2010). The Sahana emergency response software platform was used by various NGOs to organize their response to the emergency (Nelson, Sigal, and Zambrano 2010). Traditional aid agencies like the US military used a form of wiki software to help alleviate bottlenecks in their relief efforts by allowing their members to contribute information about the disaster that responders could use to make more effective decisions (Yates and Paquette 2011).

The various VGI platforms employed during the Haiti earthquake illustrate how the technology can be used to meet the needs of emergency responders through real-time data access and for baseline data creation. For instance, in areas where there are few resources available for critical needs like water and shelter, VGI can create the necessary baseline data that first responders need for allocating those resources. VGI can provide real-time data on the conditions of victims and infrastructure that may take days or weeks for authoritative sources to develop (Goodchild and Glennon 2010). The data created by VGI can be open source and free, resulting in data-sets that are stored in formats that are non-proprietary and can be used by anyone. VGI also helps meet the emotional needs of affected people by transforming them from powerless victims to empowered citizens (Elwood 2008a, 2008b).

While some researchers see VGI as a benefit because of its ability to democratize access to, and creation of geospatial data (Elwood 2008a, 2008b), others worry about the quality of the data that are being produced by non-professionals (Goodchild and Glennon 2010). This concern is valid because of the important role that geospatial data provides in DSSs, especially related to emergency response when lives are at risk.

Data quality

Traditional measures of spatial data quality were appropriate when data were collected and distributed by only a few agencies, but those same measures may not be well suited to assessing VGI data (Goodchild 2008a, 2008b). The quality of geospatial data is an important consideration because of the role that maps and spatial data play in the decision-making process (Chrisman 1991; Foody 2003). Though most people initially think of positional accuracy when concerned with geospatial data quality, quality is actually comprised of many different components including lineage, positional accuracy, attribute accuracy, logical consistency, completeness, semantic accuracy, usage, purpose and constraints, and temporal quality (Van Oort 2006). *The component of data quality most relevant to this research is attribute accuracy.* Girres and Touya (2010) list three separate components of attribute accuracy: quantitative accuracy, non-quantitative attributes, and classification of features.

Various techniques for assessing attribute accuracy are being used on VGI. The most common type of VGI analyzed is OpenStreetMap (OSM) data. Through these measures, certain trends in the OSM data-set were revealed. For example, the quality of the OSM data-set improves with increasing population density and socioeconomic status and decreases in areas of low-density and socioeconomic status (Girres and Touya 2010; Haklay et al. 2010; Haklay and Ellul 2010; Zielstra and Zipf 2010). In urban areas with relatively high socioeconomic

status, the positional accuracy, completeness, and attribute accuracy rivals that of more authoritative sources, but changes to a more heterogeneous quality as the distance from urban areas increases (Haklay et al. 2010; Haklay and Ellul 2010; Zielstra and Zipf 2010).

While this research is beneficial because it illustrates that data created through crowdsourcing and VGI can be nearly as accurate as authoritative data-sets, it does not address the unique situation of data created for time-critical emergencies like disasters. There are numerous methods for assessing the quality of geospatial data, and the choice of methodology depends on the purpose of the research and the types of data available to analyze. Positional accuracy and completeness are easily compared as long as two data-sets exist for the same phenomenon, and one of those data-sets is considered more accurate than the other. Analysis methods become more complicated, however, when only one data-set exists. Using a comparative approach, this study assesses the uncertainty of attribute data produced by volunteers in response to the Haiti earthquake of January 2010.

Methods

Data collection process

Within a matter of hours after the earthquake struck Haiti, the Ushahidi Haiti web site had been established by volunteers in Boston, Massachusetts. These volunteers began collecting data from Twitter, e-mail, and traditional media sources and georeferenced this data on a web-based map of Haiti. Four days later, a SMS code 4636 was established and advertised on local Haitian radio as a way to report emergency needs. Once the 4636 SMS was established, text messages became the predominant source of data for the web site [see Morrow et al. (2011) for a detailed timeline of events and Meier (2012)¹ for a firsthand account of the creation and motivation for deploying the Ushahidi web site described in note 1].

The Ushahidi data-set was created by three sets of interconnected participants. Victims of the disaster comprised the majority of the first group and produced much of the raw VGI data that were incorporated into the Ushahidi platform. Off-site volunteers comprised the majority of the second group. Volunteers were responsible for turning the raw data produced by the victims into the information to be used by the third group – the relief agencies. This group was primarily responsible for providing aid to the victims, using the information provided on the Ushahidi Haiti web site.

An emergency response action began with a victim submitting a text message (for example) to the SMS 4636 from a cell phone. Volunteers then took this raw message and (1) translated the message from Haitian Creole to English; (2) categorized the message contents with regard to the type of emergency; and (3) georeferenced the

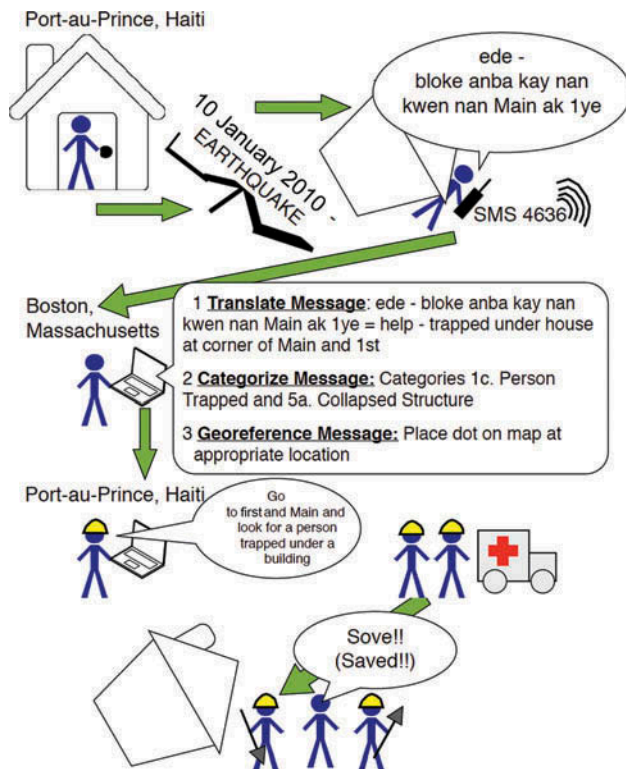


Figure 1. Ushahidi Haiti flowchart.

message to a location in Haiti. The information was then entered into the Ushahidi Haiti map database, and a symbol showing the location appeared on the web site's map, where it was then accessible to emergency responders. Figure 1 illustrates this process.

Relief agencies were then able to respond to individual messages (i.e., emergencies) by clicking on a location on the map (shown as a dot symbol), which then displayed information relevant to the emergency need. For example, the map would display the translated message, the nature/type of emergency, the time the message was posted to the web site, and the latitude and longitude of the emergency need. Figure 2 is an example image of the Ushahidi Haiti web site. The Ushahidi Haiti web site was endorsed by the US Secretary of State and used by various organizations and relief agencies including but not limited to the US Department of State, the US Marine Corp., the US Coast Guard, NYC Medics, the Red Cross, United Nations, and 'on-the-ground' Search and Rescue Teams (Morrow et al. 2011).

Categorization of emergency types

The original database was a comma separated values (CSV) formatted file downloaded from the web site <http://haiti.ushahidi.com> in Spring 2012. The database consisted of 10 columns and 3606 rows. The attributes in the database are included in Table 1. See Table 2 for an example of the database:

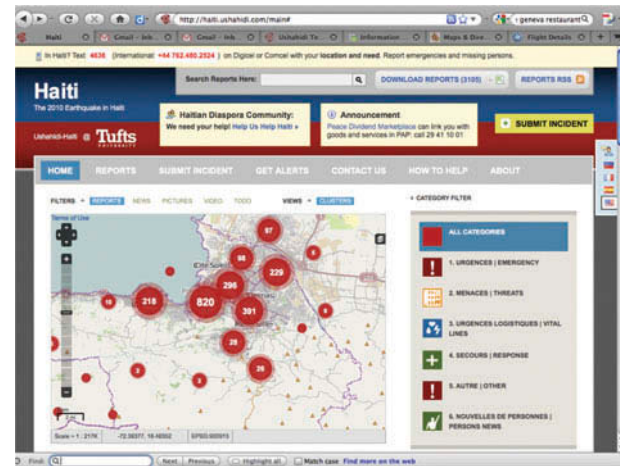


Figure 2. Ushahidi Haiti web interface (Edublog, 2011).

Table 1. Database attributes.

#	Unique integer identifier for each record
INCIDENT TITLE	Title given to each record in the database by the volunteer as determined from the original victim's message
INCIDENT DATE	The date and time stamp for when the record was added to the database
LOCATION	A description of the location referenced in the original message from the victim (e.g., the name of a specific community within a town or the nearest landmark)
DESCRIPTION	Contains the contents of the message submitted by the victim
CATEGORY	Contains any emergency type, preceded by category number, in the message submitted by the victim. Emergency categories (e.g., 1) and subcategories (e.g., 1b) were selected by the volunteers from predetermined list of emergency types
LATITUDE and LONGITUDE	The geographic coordinates of the record in WGS84 decimal degrees. The Ushahidi software generated these attributes automatically when the volunteer added a dot to the map where they thought the victims' reports were located
APPROVED	The moderators have the ability to approve a message based on whatever criteria have been established
VERIFIED	The moderators document if a message has been corroborated

Data preparation

The initial database contained 3604 records from messages received from 12 January 2010 through 7 September 2010. This study focused on text messages, twitter posts, and news web site broadcasts received during the first two weeks post disaster, as this closely

Table 2. Sample Ushahidi Haiti CSV.

#	Incident Title	Incident Date (Time)	Location	Description	Category	Latitude	Longitude	Approved	Verified
638	Anesthesiologist needed	17 January 2010 (02:08:00)	Port-au-Prince	In the Hospital de la Paix are anesthesiologists and other doctors needed	1. Urgences Emergency, Ib. Urgence medicale Medical Emergency, 7. Secours Services Available	18.556439	-72.298248	Yes	No
4054	Pharmacy open	27 July 2010 (20:20:00)	Lamarre	Pharmacy at Lamarre Street in Petionville open and selling medications		49.295769	-0.892294	No	No
4051	Food aid sent to Fondwa, Haiti	28 June 2010 (23:06:00)	Fondwa	Please help food aid. organizations deliver more food to Haiti through your financial gift and donations.	1. Urgences Emergency, 2. Urgences logistiques Vital Lines	50.226029	5.729886	No	No

coincides with the official end-of-rescue operations in Haiti (Batty 2010; BBC News 2010). This period post disaster (12–27 January) included 2608 records to be considered for analysis. Of these 2608 records, 28 were not in English (had not been translated from Haitian Creole) and were therefore excluded from analysis. Of the remaining 2580 records, 90 were duplicates in the database and were excluded from the analysis. The result was an analysis of 2490 records (95.5% of 2608 records).

Reclassification of text messages

To assess the consistency of how volunteers classified the content (i.e., emergency type) into primary categories and subcategories of emergency types, we conducted an independent reclassification of the 2490 messages. This entailed reading each message and assigning a primary category and subcategory of each emergency need expressed in a message. See Tables 3 and 4 for a list of all primary and subcategory emergency types in the original data-set. A quantitative comparison of the classification produced by the volunteers against the results of the independent reclassification was conducted.

To perform a reclassification consistent to the volunteers', the volunteer community was contacted via e-mails, and categorization guidelines including class definitions and training materials were requested. It was discovered that guideline documents did not exist to instruct volunteers (J. Valuch, personal communication, 15 January 2013). Subsequently, definitions and guidelines for primary and subcategories were created to at least provide consistency in the reclassification that was performed for this study (see Appendix).

All references to the original categories and subcategories were removed from the 2490 records in order to reduce bias in the reclassification. The 2490 records were then reclassified. A second independent reclassification was then conducted as a check for consistency. Records whose reclassification did not match were checked again. The two narratives below are examples of how reclassification was performed:

Table 3. Main categories in Ushahidi Haiti deployment.

Primary Category Number	Emergency Type
1	Emergency
2	Vital lines
3	Public health
4	Security threats
5	Infrastructure damage
6	Natural hazards
7	Services available
8	Other

Record # 516 (example of agreement in classification)

Content of Message: *Carrefour, Fontamura, Bizotton, Thor: Hopital Adventiste de Diquini (Haitian Adventist Hospital) is treating and receiving patients in and around Carrefour (Fontamara, Bizotton, Thor, etc.).*

Original Category By Volunteers:

- 7d. Hospital/clinics operating

Reclassification:

- 7d. Hospital/clinics operating

Category Definition Developed for this Study:

- Report that medical services are being provided at this location

Justification:

- Message specifically mentions that the hospital is treating and receiving patients.

Record #527 (example of disagreement in classification)

Contents of Message: *People trapped under house screaming (address and name withheld for privacy)*

Original Category By Volunteers:

- 1c. People trapped

Reclassification:

- 1c. People trapped; 5a. Collapsed structure

Category Definition Developed for this Study:

- 1c. reports of people trapped who are still alive, must be specific about location; 5a. reports indicating collapsed buildings that are specific with a building/company name or address

Justification

- The message clearly indicates that people are trapped so that relates to category 1c. Because the message also uses 'under house' then that implies that the structure has collapsed and therefore relates to category 5a.

Table 4. Subcategories for Ushahidi Haiti deployment.

Category Number	Title	Category Number	Title
1a	Highly vulnerable	5a	Collapsed structure
1b	Medical emergency	5b	Unstable structure
1c	people Trapped	5c	Road blocked
1d	Fire	5d	Compromised bridge
2a	Food shortage	5e	Communication lines down
2b	Water shortage	6a	Floods
2c	Contaminated water	6b	Landslides
2d	Shelter needed	6c	Earthquakes and aftershocks
2e	Fuel shortage	7a	Food distribution point
2f	Power outage	7b	Water distribution point
3a	Infectious human disease	7c	Non-food distribution point
3b	Chronic care needs	7d	Hospital/clinics operating
3c	Medical equipment and supply needs	7e	Feeding centers available
3d	OBGYN/Women's Health	7f	Shelter offered
3e	Psychiatric need	7g	Human remains management
3f	Water sanitation and hygiene promotion	7h	Rubble removal
3g	Deaths	8a	IDP concentration
4a	Looting	8b	Aid manipulation
4b	Theft of aid	8c	Price gouging
4c	Group violence	8d	Search and Rescue
4d	Riot	8e	Person news
4e	Security concern	8f	Other
		8g	Missing persons
		8h	Asking to forward a message

For the reclassification, binary fields were created for each category and subcategory and were then populated based on the results of the categorization. This categorization resulted in eight main categories and 46 subcategories. The cause of the difference in the number of subcategories between the original database and the reclassified database deserves some attention. An analysis of the contents of each record suggests that volunteers might not have applied subcategorization consistently. When the volunteer community was queried as to how subcategories were used and applied during the post-disaster period, it appears no documentation existed (J. Valuch, personal

communication, 15 January 2013). The result is that in some instances, the same subcategory had two definitions. See Table 5 for examples.

In other instances, subcategories were moved to more closely match their more appropriate main category. See Table 6 for examples.

To facilitate comparison of the original categorization of 2490 messages in the original database to the reclassification performed as part of this study, a Python script was used. A spreadsheet software program was then used to generate descriptive statistics to determine how many records were a match between the two data-sets at both the category level and the subcategory level. Table 7 illustrates the output of the Python script, as well as a description of what the columns represent. Referring to the example in Table 7, the original volunteers determined that record 67 (column 1) in the database related to categories 1, 7, and 8 (column 2). When re-categorized, it was

determined that that message only related to category 8 (column 3). The Python script then determined that for record 67, the original categorization and the reclassification both related to category 8 (column 4). The last two columns indicate which categories appeared in only the original volunteer categorization and the re-categorized data-set. For record 68, there was agreement that the message related to category 6 (column 4), while the volunteers thought the message also applied to category 7 and the reclassification placed it in category 8. For record 69, there was no agreement between the original classification and the reclassification.

Results and analysis

Main categories

The aggregate results for the main categories show that the number of differences in entries for each category and the total number of entries for each category varied widely across categories and subcategories (see Table 8).

Using Blalock's difference of proportion test, category 1 (Emergency) had significantly fewer entries after reclassified. It comprised 13.98% of the total original entries and 6.64% of the total reclassified entries ($Z = 2.87$, $p = 0.004$). Category 2 (Vital Lines) contained the highest percentage of the total records in both the original data-set and the reclassified data-set, accounting for 39.57% of the former and 44.18 of the latter, which was again a significant difference ($Z = 2.52$, $p = 0.012$). Category 3 (Public Health) had a significant increase in the number of entries after reclassification ($Z = 2.51$, $p = 0.012$). It comprised 6.34% of the total original entries and 12.66% of the re-categorized entries. While category 4 (Security Threats) had fewer entries after reclassification, with 5.80% of the volunteer categorization and only 1.80% after reclassification, this difference was not significant ($Z = 1.28$, $p = 0.201$). Category 5 (Infrastructure Damage) had moderately significant more entries after reclassification, comprising 4.86% before and 9.17% after ($Z = 1.71$, $p = 0.089$). Category 6 (Natural Hazards) did not have a significant difference in the number of entries when comparing the original categorization to the reclassified data, comprising 1.08% of the total before and 1.22% after

Table 5. Subcategories with more than one definition.

Subcategory	Definition 1	Definition 2
2c	Contaminated water	Security concern
6a	Floods	Deaths
6b	Landslides	Missing persons
6c	Earthquakes and aftershocks	Asking to forward a message

Table 6. Subcategories moved to different main categories.

Original Subcategory and Definition	Original Category	New Category	New Subcategory
2c – Security concern	2 – Vital lines	4 – Security threats	4e
4e – Water sanitation	4 – Security threats	3 – Public health	3f
6a – Deaths	Natural hazards	3 – Public health	3g
6b – Missing persons	Natural hazards	8 – Other	8g
6c – Asking to forward a message	Natural hazards	8 – Other	8h

Table 7. Sample CSV generated by Python script comparing original and re-categorized results per record.

1	2	3	4	5	6
Unique ID	Original Categories (Each Digit Represents a Category)	Reclassified Categories	Agreement Between Classifications	In Original Data-Set Only	In Reclassified Data Only (Each Digit Represents a Category)
67	178	8	8	17	
68	67	68	6	7	8
69	125	8		125	8

Table 8. Instances of each main category in total.

Category	Number of Entries in Each Category for the Original Data-Set	Number of Entries in Each Category After Reclassification	Z-Score (Blalock's Difference of Proportions Test)	Probability
1	492	228	2.856	0.004
2	1393	1518	-2.515	0.012
3	223	435	-2.513	0.012
4	204	62	1.279	0.201
5	171	315	-1.707	0.077
6	38	42	-0.060	0.952
7	588	223	3.755	0.002
8	411	613	-2.681	0.007
Total	3520	3436		

Notes: The total entries for both the original data-set and the re-categorized data-set are higher than the 2490 text messages. This is due to some messages containing more than one emergency category and/or subcategory. Bold values are significant at $p < 0.05$ and italic values are marginally significant at $p < 0.1$ and ≥ 0.05 .

($Z = 0.06$, $p = 0.952$). Category 7 (Services Available) had significantly fewer entries after reclassification, comprising 16.70% of the original entries and only 6.49% after reclassification ($Z = 3.755$, $p = 0.002$). Note that category 7 had the second most entries in the original data-set, but the third least entries in the reclassified data-set. Category 8 (Other) had significantly more entries after reclassification than before, comprising 11.68% of the total original entries and 17.84% after reclassification ($Z = 2.68$, $p = 0.007$).

Subcategories

The number of entries within each subcategory both for the original data-set for and the revised data-set is highly variable. Even within category 2, the largest category both before and after reclassification, there were some individual subcategories with very few entries. Some of this is attributed to omission and commission inconsistencies between the original and revised data-set, while others are due to rearrangement of some subcategories. Rearranging of subcategories will be highlighted below as appropriate. Again using Blalock's difference of proportions test, only two of the subcategories had significantly different numbers of entries after reclassification (2b, $Z = 2.57$, $p = 0.01$; and 2d, $Z = 2.22$, $p = 0.026$). Many of the subcategories could not be analyzed for differences in classification because the samples were either too small, or zero.

Category 1 consisted of four subcategories related to emergencies (Table 9). Subcategory 1d (*Fire*) had very few entries both before and after reclassification but had

Table 9. Subcategories: The fourth column indicates the difference between subcategory classification by the volunteers and subcategory reclassification.

Subcategory	Number of Entries in Each Subcategory – Classification by Volunteers	Number of Entries in Each Subcategory after Reclassification	Difference Between Classification by Volunteers vs. Reclassification
1a	0	6	-6
1b	170	61	109
1c	155	157	-2
1d	4	4	0
2a	998	942	56
2b	879	667	212
2c	17	3	14
2d	183	394	-211
2e	17	36	-19
2f	9	4	5
3a	0	4	-4
3b	0	1	-1
3c	222	267	-45
3d	0	1	-1
3e	0	2	-2
3f	0	26	-26
3g	0	53	-53
4a	17	13	4
4b	0	0	0
4c	0	0	0
4d	0	1	-1
4e	151	47	104
5a	129	258	-129
5b	28	26	2
5c	27	22	5
5d	0	2	-2
5e	0	5	-5
6a	2	0	2
6b	15	2	13
6c	21	6	15
7a	210	23	187
7b	2	23	-21
7c	50	20	30
7d	217	128	89
7e	0	3	-3
7f	0	7	-7
7g	29	5	24
7h	7	0	7
8a	0	185	-185
8b	0	4	-4
8c	0	15	-15
8d	44	4	40
8e	275	15	260
8f	0	186	-186
8g	0	215	-215
8h	0	2	-2

Note: Negative values indicate when reclassification resulted in more observations within that subcategory.

the exact same number of entries in both cases. Subcategory 1a (*Highly Vulnerable*) had no entries by volunteers, but six entries in the reclassified data-set. Subcategories 1b (*Medical Emergencies*) and 1c (*People*

Trapped) comprised the majority of the entries both before and after reclassification. Category 2 consisted of six subcategories related to needs. Subcategories 2a (*Food Shortage*) and 2b (*Water Shortage*) comprised the overwhelming majority of the entries both before and after reclassification.

Category 3 was originally comprised of five subcategories related to public health; however, after examining all of the data, two subcategories were reassigned to this category (see Table 6). Original category 4e (*Water Sanitation*) became 3f and subcategory 6a (*Deaths*) became 3g. Subcategories 3a, 3b, 3d, and 3e contained no entries in the original data-set and a very limited number of entries in the revised data-set. Category 3c (*Medical Equipment and Supply Needs*) comprised the overwhelming majority of entries in both the original and the revised data-sets. Due to subcategories 3f and 3g being moved into category 3 during reclassification, they cannot be compared to classification done by volunteers.

Category 4 originally consisted of five subcategories related to security threats. Subcategory 4e (*Water Sanitation*) was moved to category 3 and replaced with subcategory 2c. (In the original data-set, there were two 2c subcategories – one referenced *Contaminated Water* while the other referenced *Security Concern*.) Original subcategories 4b, 4c, and 4d did not contain any entries in the original data-set and only 4d (*Riot*) contained an entry in the revised data-set.

Category 5 consisted of five subcategories related to infrastructure damage. The majority of entries in both the original and the revised data-sets related to subcategory 5a, *Collapsed Structure*, followed by 5b, *Unstable Structure*, and 5c, *Road Blocked*. There were no original entries in subcategories 5d and 5e and only a few in the revised data-set. Category 6 was a challenging category. The original data-set consisted of two definitions each for category 6's three subcategories, which were supposed to relate to natural hazards. As a result, the second 6a (*Deaths*) was moved to category 3, the second 6b (*Missing Persons*) was moved to category 8, and the second 6c (*Asking to Forward a Message*) was moved to category 8. The remaining three subcategories related to natural hazards. The overall number of entries for category 6 in both the original and revised data-sets was very small with no instances of subcategory 6a in the revised data-set.

Category 7 consisted of eight subcategories related to services available. The two largest subcategories in both the original and revised data-sets were 7a, *Food Distribution Points*, and 7d, *Hospital/Clinics Operating*. There were no original entries for subcategories 7e, *Feeding Centers Available*, and 7f, *Shelter Offered*, and there were no revised entries for subcategory 7h, *Rubble Removal*.

Category 8 originally consisted of six subcategories that did not fit in any other primary category. After the second subcategories 6b and 6c were moved, however, category 8 consisted of eight subcategories. Four of the original subcategories did not contain any entries, with most of the original entries going to subcategory 8e, *Person News*. Somewhat surprisingly, there were no original entries for subcategory 8a, *IDP (Internally Displaced Persons) Concentrations*.

Independent Ushahidi Review

Table 10 reports the results (column 2 in the table) of an independent review of the Ushahidi deployment in Haiti approximately a year and a half after the earthquake conducted by Morrow et al. (2011). This analysis was done by randomly selecting 50 entries from the total of 3584 and assessing their original categories for errors of omission, commission, or a combination of omission and commission. Using these randomly selected entries, the reviewers estimated an overall error rate of 36% with a perfect match rate of 64%. In comparison (column 3), this study determined an overall error rate of 50%.

Table 11 contains the results for the study period at the subcategory level that most closely matches the analysis done by the independent reviewers. This is a much more stringent level of assessment because it checks for agreement among all of the subcategories. Note that the overall error rate was determined to be 73%.

Table 10. Results of independent Ushahidi evaluators vs. the results of this study.

Error Type	Percent Error In Classifications Assessed By Independent Review	Percent Error in Classifications Assessed in this Article
Reports with incorrect category tag (commission)	18	123
Reports missing a critical category tag (omission)	30	15
Both incorrect and missing tags	6	23
Missing or incorrect category tag (overall error rate)	36	50
Reports with neither missing nor incorrect tags	64	50

Note: A statistical comparison was not reliable due to the small sample size (50) assessed by the Ushahidi independent review compared to this study (3584).

Table 11. Category assessment by discrepancy type, reclassified data, subcategory level

Error Type	Percent of all Reports
Reports with incorrect category tag (commission)	17
Reports missing a critical category tag (omission)	24
Both incorrect and missing tags	17
Missing or incorrect category tag (overall error rate)	73
Reports with neither missing nor incorrect tags	27

Discussion

Based on the apparent attribute classification error and inconsistencies in the original data-set produced by the volunteers compared to the reclassified data-set we created, it would be easy to dismiss VGI as a viable data source during disasters. However, these differences do not tell the entire story of this data-set and its significance. Prior to the earthquake that struck Haiti in early 2010, almost all data produced during disasters came from authoritative sources. Authoritative data had a tendency to be slow to produce and was often not available to anyone outside the disaster response community. In addition, the collected data were often derived from remote sensing products, or a limited number of data collectors on the ground. This greatly limited the amount of data produced and the types of data that could be collected. In contrast to this historical approach to data collection, the volunteer response to the Haiti disaster produced a major paradigm shift in emergency response.

In a way, the Ushahidi deployment in Haiti served as a national 9-1-1 system that was completely operated by volunteers. The amounts and types of data available to responders were significantly increased. Volunteers were able to collect data at a very fine scale from individual victims. Volunteers were also able to collect data that are not readily visible like hunger, thirst, and price gouging that may be missing from remotely sensed data. The data that were collected by volunteers were immediately available rather than having to wait to be incorporated into the next map update. The data were also available in an open format to anyone interested in the disaster. This helped reduce many of the barriers present in traditional emergency response situations when dealing with incompatible software or data or when trying to navigate licensing agreements.

In addition, the benefit of this Ushahidi database to relief agencies is critical to recognize (Meier 2012). On 22 January, The US Coast Guard contacted Meier to enquire how 'better to use the information in Ushahidi':

I am with the US Marine Corps. I am stateside assisting the 22 MEU [Marine Expeditionary Unit] coming off the USS Bataan [on the Haitian Coast]. We want to use your data to bring aid to the people Haiti right now. The USMC is focusing on Leogane, Grand Goave, and Petit Goave. Is there a way to import your data into Google Earth or GIS? We want to make this work for the people of Haiti... please let me know ASAP (Meier 2012)

Subsequently, a dedicated 'Skype chat was created with the Coast Guard to fast forward the most urgent (and actionable) content that was being added to the live Haiti Crisis Map'. The American Search and Rescue team requesting, 'GPS coordinates for the locations of trapped individuals', also contacted Meier and colleagues. FEMA's Administrator Craig Fugate tweeted: 'Crisis Map of Haiti represents the most comprehensive and up-to-date map available to the humanitarian community'. A last example of the many benefits of this Haiti Ushahidi Crisis Map comes from the US Marine Corps:

I can not overemphasize to you what the work of the Ushahidi/Haiti has provided. It is saving lives every day. I wish I had time to document to you every example, but there are too many and our operation is moving too fast. Here is one from the 22 MEU [Marine Expeditionary Unit]: 'We had data on an area outside of Grand Goave needing help. Today, we sent an assessment team out there to validate their needs and everything checked out. While the team was out there, they found two old women and a young girl with serious injuries from the earthquake; one of the women had critical respiratory issues. They were evacuated'.

Your site saved these people's lives. I say with confidence that there are 100 s of these kind of stories. The Marine Corps is using your project every second of the day to get aid and assistance to the people that need it most. We did have a tech barrier that we had to surmount. The Marines downrange have Google Earth and your site does not work on the ship for them. So, I had Georgia Tech create a bridge from your site to Google Earth.

But it is YOUR data and YOUR work that is putting aid and assistance directly on the target and saving lives. Our big gap right now is locating NGOs and where they are working. Your site is helping with that. Keep up the good work!! You are making the biggest difference of anything I have seen out there in the open source world. (Meier 2012)

In spite of these benefits, however, the use of VGI data produced using the Ushahidi platform has room to improve. The following ideas are meant to stimulate discussion about ways to improve the quality of data produced as part of the Ushahidi platform. Because this study revolved around attribute consistency in the data, the suggestions presented here focus on techniques that may improve that consistency.

First, organizers of the online crisis mapping community should consider developing documentation and training materials to enable volunteers to be more succinct in

classifying emergencies. For example, training materials could provide predefined categories and subcategories, complete with definitions and examples. Members of the online crisis mapping community should consider other hierarchical systems for classification, like the USGS Land Cover System, to establish a more uniform and widely used system. Organizers should consider creating a process for documenting changes in procedures used during an emergency. For instance, based on data in the original CSV, it appears that the categories and subcategories changed at least once during the first few weeks after the earthquake in Haiti. When the volunteer community was queried as to what those original categories were and when they were changed, no one was able to answer that question (J. Valuch, personal communication, 15 January 2013).

Other recommendations that could be generalizable to all VGI for emergency response include volunteers using a character encoding that is appropriate for the language of the victims. Numerous reports in the original CSV were difficult to read because of what appeared to be missing or inappropriate characters. Numerous reports described a need or activity at more than one location. Volunteers should determine if a system for splitting messages might be useful in the future if a message contains references to more than one location. There were several instances of roads being closed due to debris or landslides and then follow-up reports that said those same features were now open. These different messages provided conflicting information. Volunteers should consider whether adding an ability to change the status of a report is needed. Many reports in the original CSV were meant as ways to collect data directly from victims. However, numerous reports were also published that related to information sharing between relief agencies. It may be worth investigating whether a multiple-tiered system could be appropriate so that data can be collected from victims at one level, and that relief agencies can coordinate with each other at another level.

Many messages from victims voiced a great deal of frustration at having reported a need but not receiving help. In the future, it may be beneficial to help manage the expectations of the victims so that they do not feel neglected when they are already in a vulnerable situation. While some messages were quite specific about the types of aid that were needed and where the victims were located, many other messages were quite generic with people just asking for help. It may also be worthwhile to investigate ways to better communicate what information is needed from the victims in order to act on their reports.

There were many reports that matched a main category but did not suite any of the appropriate subcategories. It might be worth considering adding an *Other* subcategory to each main category rather than having a single main

category labeled *Other*. For instance, there were several instances where people requested help with corpse removal. This type of request may best fit in the main category related to public health, but because there was not a specific subcategory for corpse removal, they were placed in the other main category.

Conclusions

This research project sets out to better understand the consistency of emergency attribute classification by volunteers in a time-critical emergency. An independent review of the Ushahidi Haiti deployment estimated that 64% of the reports in the original database were correctly categorized (Morrow et al. 2011). After re-categorizing the data, this study found that only 27% of the reports in the original database were correctly categorized at the subcategory level. However, aggregation of subcategories to their main category found that 50% of the messages were appropriately categorized. In other words, categorization at a coarser level resulted in greater accuracy than finer-detailed classification. While nearly 50% accuracy at the main category level is a significant improvement over 27% at the subcategory level, it is still considerably less than the 64% accuracy estimated by the independent reviewers of the Ushahidi Haiti data-set.

We should consider that there are several potential limitations related to the methods used in this study. While we believe we were diligent conducting our reclassification, we realize that our reclassification of messages into emergency types could be biased by our prior knowledge, and by our understanding of the original classification conducted by the Ushahidi volunteers. In addition, when it was discovered that training materials to guide the volunteers' classification efforts did not exist, we had to develop our own definitions of categories, resulting in potential bias.

Considering more traditional methods of comparing attributes, it is common to compare a data-set of unknown quality to one of known quality (Goodchild and Hunter 1997). However, no authoritative data-set existed that was suitable to compare against the original Ushahidi data-set, so we devised our own. Our reclassified data-set is an example of constructing an authoritative data-set to be used for comparison.

Another limitation of this study is that while it addressed inconsistencies in classification of emergency attributes, we were unable to investigate the causes because those causal data either were not available, or do not exist. In addition, we investigated a single Ushahidi deployment – subsequent deployments that took place after the earthquake in Haiti might have significantly different results. Last, this study did not address the effects of error in the attribute data, and whether or not they had any impact on relief activities.

Future research

Future research should be conducted as a way to further explore the data-sets used in this study. For example, one might examine any trends in inconsistencies between the original data and the revised data over time – did the robustness of volunteer classification change over the two-week period? While it is beneficial to know the magnitude of the discrepancies between the two data-sets, further studies could study the nature or cause of the discrepancies. Additional studies might consider other components of geospatial data quality related to this disaster. Longitudinal studies could be conducted to determine if the discrepancies are consistent across multiple deployments of the Ushahidi platform.

Additional research could measure the direct benefits to individuals or to organizations that utilize the Ushahidi platform. For instance, what are some ways to incorporate the strengths of VGI in disaster response to supplement data produced by traditional aid agencies? GIS models could be developed to help predict the accuracy of information collected by volunteers to help responders assess the appropriateness of a data source. Sensitivity analyses could also be conducted to determine which aspects of geospatial quality have the greatest impact on activities undertaken by emergency responders.

Beyond the specific research ideas mentioned above that relate specifically to this thesis, we believe that there are several broad categories that deserve further attention. An important component of future VGI and disaster response research should focus on the equality of access to the tools needed to create and share the data. Haklay and Ellul (2010) present evidence to suggest that the producers of VGI do not span all socioeconomic segments of society, but rather skew away from people who are at the low end of the socioeconomic spectrum. This could prove dangerous because these same people may be most at risk of being unable to evacuate prior to an emergency and may not have access to the resources necessary to survive and rebuild. Future research may need to focus on making the tools necessary to create and share VGI more equally ubiquitous.

Another important component of future VGI research is the degree that data creation is affected by the disaster itself. Nelson, Sigal, and Zambrano (2010) report that, despite the unprecedented role of crowdsourcing and VGI in response to the Haiti earthquake, the use of cell phones was reliant on damaged and overtaxed cellular networks and access to electricity to charge and use the devices, while inexpensive, battery powered, low-tech FM radio was available throughout the duration of the earthquake. It may prove that, despite the allure of new technology like location-aware cell phones and web applications, a combination of low and high tech efforts may be the best way to collect and provide information

following a natural or human-made disaster that disrupts critical infrastructure. Future research could focus on combining technology that is more resistant to damage from disasters with newer technology that may or may not be available due to damage.

Future research could also focus on how people actually interact with social media tools, especially in high-stress situations (American Red Cross 2010). Will people be more likely to use social media tools during an emergency if they use them in their normal life? Will the data they provide be more useful if they are already familiar with how the tools work rather than trying to learn them in a high-stress situation? The same research could be applied to those institutions and agencies that may want to incorporate VGI into their DSS. If these agencies are able to test and experiment with VGI during normal operations, will they be more likely to use VGI during disasters? Also, will they make better use of the data if they already have a system in place, rather than waiting until an emergency to learn?

A combination of law, politics, and geography may also present a new research agenda related to VGI. Are relief agencies required to act on data that are submitted through social media like Twitter? What if the data are submitted anonymously? Will they be held liable if they fail to act even if responding uses limited resources? Will society provide the funding and resources to provide relief agencies with the new tools and training to take advantage of social media and VGI?

As the cost of data acquisition continues to decline, the role of VGI in society will continue to grow. Increasingly people have access to cell phones and the Internet. The capabilities of those cell phones continue to expand, and even when they do not provide smart phone capabilities, they can be used to provide useful information (Munro 2010). The adoption of social media applications like Facebook and Twitter shows no signs of slowing, and as they continue to gain new users and features, their role in society will continue to grow and provide researchers with ever more data to mine. Relief agencies like the United Nations have begun to recognize the importance of VGI in their operations and are conducting their own research into VGI's usefulness and applications (Standby Volunteer Task Force 2011).

Note

1. Patrick Meier at Harvard began the Ushahidi mapping for Haiti. As the volume of tweets increased, he enlisted friends to help process the tweets. The volume of information continued to increase from multiple sources included tweets, social media, and traditional media, therefore 100 graduate and graduate students were enlisted from The Fletcher School at Tufts University (where Meier received his Ph.D.) to help monitor and map information. Mapping was

difficult because of the paucity of geospatial data available in digital format, so OpenStreetMap volunteers used World Band supplied satellite imagery to trace roads. The short message system for Haitian Diaspora was created, resulting in thousands of messages – most were in Haitian Creole and needed to be translated. Additional volunteers from Haitian Diaspora in Boston translated the messages to English and helped to locate the messages on the map. So many messages were received that the graduate and undergraduate student volunteers triaged messages to focus on most serious life and death messages (Meier 2012).

References

- American Red Cross. 2010. "Social Media in Disasters and Emergencies." Accessed December 3, 2011. www.american-redcross.org.
- Batty, D., 2010. "Haiti Ends Quake Rescue Operations." Accessed May 29, 2014. <http://www.theguardian.com/world/2010/jan/23/haiti-ends-quake-rescue-operations>.
- BBC News. 2010. "Haiti Quake Victim Rescue Operation Declared Over." Accessed May 29, 2014: <http://news.bbc.co.uk/2/hi/americas/8476474.stm>.
- Bloch, M., and S. Carter. 2012. "Twitter Chatter During the Super Bowl." *The New York Times* 2009. Accessed April 15. http://www.nytimes.com/interactive/2009/02/02/sports/20090202_superbowl_twitter.html.
- Chrisman, N. R. 1991. "The Error Component in Spatial Data." In *Geographic Information Systems*, edited by P. A. Longley, M. F. Goodchild, D. J. Maguire, and D. W. Rhind. London: Longman Scientific and Technical.
- Cova, T. J. 1999. "GIS in Emergency Management." In *Geographical Information Systems: Principles, Techniques, Applications, and Management*, edited by M. F. G. P. A. Longley, D. J. Maguire, and D. W. Rhind. New York: John Wiley & Sons.
- Curran, P., C. De Silva, and B. Van De Walle. 2007. "Open Source Software for Disaster Management." *Communications of the ACM* 50 (3): 61–65. doi:10.1145/1226736.1226768.
- Edublogs. 2011. "Ushahidi-Haiti." Last modified April 11. Accessed June 5, 2013. <http://kristenm051.edublogs.org/files/2011/04/Screen-shot-2011-04-11-at-8.42.46-AM-uyf7we.png>.
- Elwood, S. 2008a. "Volunteered Geographic Information: Future Research Directions Motivated by Critical Participatory, and Feminist GIS." *GeoJournal* 72: 173–183. doi:10.1007/s10708-008-9186-0.
- Elwood, S. 2008b. "Volunteered Geographic Information: Key Questions, Concepts and Methods to Guide Emerging Research and Practice." *GeoJournal* 72: 133–135. doi:10.1007/s10708-008-9187-z.
- Foody, G. M. 2003. "Uncertainty, Knowledge Discovery and Data Mining in GIS." *Progress in Physical Geography* 27 (1): 113–121. doi:10.1191/0309133303pp345pr.
- Freifeld, C. C., R. Chunara, S. R. Mekaru, E. H. Chan, T. Kass-Hout, A. A. Iacucci, and J. S. Brownstein. 2010. "Participatory Epidemiology: Use of Mobile Phones for Community-Based Health Reporting PLOS." *Medicine* 7 (12): 1–5.
- Girres, J. F., and G. Touya. 2010. "Quality Assessment of the French OpenStreetMap Dataset." *Transactions in GIS* 14 (4): 435–459. doi:10.1111/j.1467-9671.2010.01203.x.
- Goodchild, M. F. 2007. "Citizens as Sensors: the World of Volunteered Geography." *GeoJournal* 69 (4): 211–221. doi:10.1007/s10708-007-9111-y.
- Goodchild, M. F. 2008a. Assertion and Authority: The Science of User-Generated Geographic Content. Paper read at *colloquium for Andrew U. Frank's 60th birthday*, GeoInfo 39, Department of Geoinformation and Cartography, Vienna University of Technology, June.
- Goodchild, M. F. 2008b. Spatial Accuracy 2.0. Paper read at keynote address, 8th international symposium on spatial accuracy assessment in natural resources and environmental sciences, Shanghai, June 25–27.
- Goodchild, M. F., and J. A. Glennon. 2010. "Crowdsourcing Geographic Information for Disaster Response: A Research Frontier." *International Journal of Digital Earth* 3 (3): 231–241. doi:10.1080/17538941003759255.
- Goodchild, M. F., and G. J. Hunter. 1997. "Dealing with Error in Spatial Databases: A Simple Case Study." *Photogrammetric Engineering and Remote Sensing* 61 (5): 529–537.
- HOT. "Haiti." Accessed June 30, 2013. <http://hot.openstreetmap.org/projects/haiti-2>.
- Haklay, M. 2010. "How Good Is Volunteered Geographical Information? A Comparative Study of OpenStreetMap and Ordnance Survey Datasets." *Environment and Planning B: Planning and Design* 37 (4): 682–703. doi:10.1068/b35097.
- Haklay, M., S. Basiouka, V. Antoniou, and A. Ather. 2010. "How Many Volunteers Does it Take to Map an Area Well? The Validity of Linus' Law to Volunteered Geographic Information." *The Cartographic Journal* 47 (4): 315–322. doi:10.1179/000870410X12911304958827.
- Haklay, M., and C. Ellul. 2010. "Completeness in Volunteered Geographical Information the Evolution of OpenStreetMap Coverage in England (2008–2009)." *Journal of Spatial Information Science*. <http://povesham.wordpress.com/2010/08/13/completeness-in-volunteered-geographical-information-%E2%80%9393-the-evolution-of-openstreetmap-coverage-2008-2009/>!
- Hesse, M., 2010. "Crisis Mapping Brings Online Tool to Haitian Disaster Relief Effort." Accessed May 29, 2014. <http://www.washingtonpost.com/wp-dyn/content/article/2010/01/15/AR2010011502650.html?hpid=topnews>.
- Howe, J. 2008. *Crowdsourcing: Why the Power of the Crowd Is Driving the Future of Business*. New York: Three Rivers Press.
- Kaiser, R., P. Spiegel, A. Henderson, and M. Gerber. 2003. "The Application of Geographic Information Systems and Global Positioning Systems in Humanitarian Emergencies: Lessons Learned, Programme Implications and Future Research, Disasters." *Disasters* 27 (2): 127–140. doi:10.1111/1467-7717.00224.
- Kelmelis, J., L. Schwartz, C. Christian, M. Crawford, and D. King. 2006. "Use of Geographic Information in Response to the Sumatra-Andaman Earthquake and Indian Ocean Tsunami of December 26, 2004." *Photogrammetric Engineering and Remote Sensing* 72: 862–876.
- LaFranchi, H. 2010. "The Christian Science Monitor, 'Haiti earthquake: How a top UN official was plucked from the rubble'." Last modified January 26, 2010. Accessed June 4, 2013. <http://www.csmonitor.com/World/Americas/2010/0126/Haiti-earthquake-How-a-top-UN-official-was-plucked-from-the-rubble>.
- Liu, S. B., and L. Palen. 2010. "The New Cartographers: Crisis Map Mashups and the Emergence of Neogeographic Practice." *Cartography and Geographic Information Science: Special Issue on New Directions in Hazards and Disaster Research* 37 (1): 69–90. doi:10.1559/152304010790588098.

- Meier, P. 2012. "How Crisis Mapping Saved Lives in Haiti." Accessed July 2. <http://newswatch.nationalgeographic.com/2012/07/02/crisis-mapping-haiti/>. Last Accessed June 4, 2013.
- Morrow, N., N. Mock, A. Papendieck, and N. Kocmich. 2011. Independent Evaluation of the Ushahidi Haiti Project. Accessed June 4, 2013. <https://sites.google.com/site/haitiushahidieval/news/finalreportindependentevaluationoftheushahidihaitiproject>.
- Munro, R. 2010. "Crowdsourced Translation for Emergency Response in Haiti: The Global Collaboration of Local Knowledge." Paper read at AMTA Workshop on Collaborative Crowdsourcing for Translation, Denver, CO.
- National Research Council. 2007a. *Successful Response Starts with a Map: Improving Geospatial Support for Disaster Management*. Washington, DC: National Academies Press.
- National Research Council. 2007b. *Tools and Methods for Estimating Populations at Risk from Natural Disasters and Complex Humanitarian Crises*. Washington, DC: National Academies Press. <http://www.nap.edu/catalog/11895.html>.
- Nelson, A., I. Sigal, and D. Zambrano. 2010. *Media, Information Systems and Communities: Lessons from Haiti*. London: Knight Foundation.
- Norheim-Hagtun, I., and P. Meier. 2010. "Crowdsourcing for Crisis Mapping in Haiti." *Innovations: Technology, Governance, Globalization* 5 (4): 81–89. doi:10.1162/INOV_a_00046.
- O'Reilly, T. 2005. "What is Web 2.0: Design Patterns and Business Models for the Next Generation of Software." Accessed June 30, 2013. <http://oreilly.com/web2/archive/what-is-web20.html>.
- Okolloh, O. 2008. "Ushahidi, or 'Testimony': Web 2.0 Tools for Crowdsourcing Crisis Information." *Participatory Learning and Action* 59: 65–70.
- Pitzer, W. 2011. "Media and Communications Ecosystem." Accessed December 3. <https://knight.box.net/shared/static/y4m3iq9a9k43kyfkelm8.pdf>.
- Radke, J., T. Cova, M. Sheridan, A. Troy, M. Lan, and R. Johnson. 2000. "Application Challenges for Geographic Information Science: Implications for Research, Education, and Policy for Emergency Preparedness and Response." *URISA Journal* 12 (2): 15–30.
- Standby Volunteer Task Force. 2011. "Libya Crisis Map Deployment." Accessed December 3. <http://blog.standby-taskforce.com/libya-crisis-map-report/>.
- Turner, A. 2006. *Introduction to Neogeography*. Sebastopol, CA: O'Reilly.
- Ushahidi. 2011. "Ushahidi-Haiti." Last modified September 7, 2011. Accessed March 2012. <http://haiti.ushahidi.com>.
- Van Oort, P. A. J. 2006. "Spatial data quality: from description to application." PhD thesis, Wageningen Universiteit. 125.
- Van Westen, C. J., and P. Y. Georgiadou. 2001. "Spatial Data Requirements and Infrastructure for Geologic Risk Assessment." Paper read at workshop on natural disaster management, ISPRS Technical Committee VII, Ahmedabad.
- Yates, D., and S. Paquette. 2011. "Emergency Knowledge Management and Social Media Technologies: A Case Study of the 2010 Haitian Earthquake." *International Journal of Information Management* 31 (1): 6–13. doi:10.1016/j.ijinfomgt.2010.10.001.
- Zielstra, D., and A. Zipf. 2010. "A Comparative Study of Proprietary Geodata and Volunteered Geographic Information for Germany." 13th AGILE International Conference on Geographic Information Science, Guimarães.

Appendix Ushahidi Haiti category rules and definitions

Category 1 – Emergency = a time-critical response is necessary in order to preserve life and/or property

- 1a – Highly vulnerable = reports of victims who are especially vulnerable like children or the disabled but not necessarily in formalized settings like an orphanage or nursing home
- 1b – Medical emergency = reports of immediate, life-threatening illnesses or injuries that involve heavy bleeding, head trauma, etc.
- 1c – People trapped = reports of people trapped who are still alive, must be specific about location
- 1d – Fire = reports of fire

Category 2 – Vital lines = requests for services or goods that are necessary to sustain life, if a generic request that does not mention a specific subcategory below use 2

- 2a – Food shortage = reports requesting food or mentioning 'hungry', 'starving', or other words that indicate hunger
- 2b – Water shortage = reports requesting water or mentioning 'thirst', 'dehydration', or other words that indicate thirst
- 2c – Contaminated water = reports indicating that a water supply is not safe to drink
- 2d – Shelter needed = reports requesting tents, sleeping bags, tarps, clothing, or any other words that indicate cold, wet, damp, sleeping on street, etc.
- 2e – Fuel shortage = reports requesting fuel for generators, vehicles, or any other liquid fossil fuel purpose
- 2f – Power outage = reports indicating a lack of electricity

Category 3 – Public health = requests for services or goods that are necessary to prevent or treat illness or injury, also generic medical care – requests for medical care that are not life threatening, for instance, when someone says they are sick, ill, need a doctor; also requests for corpse removal

- 3a – Infectious human disease = reports indicating a specific illness not simply feeling ill or being sick
- 3b – Chronic care needs = reports indicating assistance needed for victims with life-long health needs
- 3c – Medical equipment and supply needs = requests for any items used for treating illnesses or injuries, including medically trained staff (doctors, nurses, specialists, etc.)
- 3d – OBGYN/Women's Health = requests for medical care that are specific to women's health or delivery
- 3e – Psychiatric need = requests for help with psychiatric problems
- 3f – Water sanitation and hygiene promotion (move to category 3 public health) = reports requesting items necessary for personal sanitation or hygiene like soap, water purification methods, toilets/latrines, etc. (formerly 4e)
- 3g – Deaths (move to category 3 public health) = reports that contain specific mentions of death at a specific location (should be a building or address) (former 6a duplicate)

Category 4 – Security threats = requests for security or reports of threats

- 4a – Looting = reports indicating looting or theft, but not of aid
- 4b – Theft of aid = reports indicating theft of aid
- 4c – Group violence = reports indicating riots or large groups intent on harm
- 4d – Riot = reports indicating riots or large groups
- 4e – Security concern (former 2c duplicate)

Category 5 – Infrastructure damage = reports of damage to infrastructure

- 5a – Collapsed structure = reports indicating collapsed buildings that are specific with a building/company name or address
- 5b – Unstable structure = reports indicating collapsed buildings that are specific with a building/company name or address
- 5c – Road blocked = reports indicating that a road is blocked by natural or manmade means
- 5d – Compromised bridge = reports indicating that a bridge has sustained damage or has collapsed
- 5e – Communication lines down = reports indicating that any form of communication has stopped functioning including radio, television, cellular telephone, landline, etc.

Category 6 – Natural hazards = reports of natural hazards

- 6a – Floods = reports of flooding
- 6b – Landslides = reports of landslides
- 6c – Earthquake and aftershocks = specific reports of earthquake or aftershocks

Category 7 – Services available = services that are available to victims or services or supplies that are available to other aid agencies for the ultimate purpose of helping victims

- 7a – Food distribution point = reports indicating that food is being distributed from this location

7b – Water distribution point = reports that water is being distributed from this location

7c – Non-food distribution point = reports that non-food/water items are being distributed from this location

7d – Hospital/clinics operating = reports that medical services are being provided at this location

7e – Feeding centers available = reports that meals (not food) are being provided at this location

7f – Shelter offered = reports that shelter is being offered at this location

7g – Human remains management = reports that human remains are being managed at this location

7h – Rubble removal = reports that rubble removal is taking place at this location

Category 8 – Other = any reports or requests that do not fit the above criteria

8a – IDP concentration = reports that people are concentrating at this location, must be more specific than a city or region, greater than 20 people

8b – Aid manipulation = reports that aid is not being distributed fairly or is being manipulated in any other way

8c – Price gouging = reports of price gouging

8d – Search and rescue = specific requests for search and rescue

8e – Person news = reports that people are safe if they do not already fit another category

8f – Other = requests for transportation, requests for tools to remove rubble without mentioning trapped people, requests for money, requests for jobs, reports of infrastructure repair, ambiguous message

8g – Missing persons (move to category 8 other) = reports indicating that someone is missing, can't be found, is not answering their phone, or has not been heard from (former 6b duplicate)

8h – Asking to forward a message (move to category 8 other) = reports asking for someone to pass on a message to someone else, but not related to requests for aid/help (former 6c duplicate)